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Department of Artificial Intelligence and Data Science

INTERNSHIP PROJECT REPORT

SEO Data Analytics Dashboard using Power BI: Performance Analysis of TACTIVE Building Construction Website (April – May 2026)

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Client	TACTIVE Building Construction

ABSTRACT

This project report presents an SEO Data Analytics Dashboard developed using Microsoft Power BI for TACTIVE Building Construction, completed during a one-month internship at Quenoxa Global Technologies. The dashboard analyzes website performance data collected from Google Search Console and Google Analytics for April and May 2026.

The dashboard contains five pages: Executive Dashboard, Monthly Performance Overview, Top Pages Analysis, Page Performance Summary, and Business Insights. Key metrics analyzed include Clicks, Impressions, CTR, Average Position, Active Users, and Bounce Rate. The most significant finding was a dramatic improvement in Bounce Rate from 60% in April to 13% in May, and CTR improved from 10.65% to 12.17%, indicating better user engagement.

Keywords	SEO, Power BI, Google Analytics, CTR, Bounce Rate, Data Visualization
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CHAPTER 1: INTRODUCTION

1.1 Background

TACTIVE Building Construction is a construction company that uses its website to attract clients searching for construction and building services online. Every month, Google Search Console and Google Analytics collect large amounts of data about how users find and interact with the TACTIVE website. However, without proper visualization, this data is difficult to interpret and use for decision-making.

1.2 What This Project Does

This project builds a five-page interactive Power BI dashboard that converts raw Google data into clear, easy-to-understand charts and tables. The dashboard helps the TACTIVE team quickly understand: how many people visit the site, which pages perform best, whether traffic quality is improving, and what actions to take to grow further.

1.3 Tools Used

Tool	Purpose
Microsoft Power BI	Dashboard design and data visualization
Microsoft Excel	Data cleaning and pre-processing
Google Search Console	Source for Clicks, Impressions, CTR, Position data
Google Analytics (GA4)	Source for Active Users and Bounce Rate data

CHAPTER 2: DASHBOARD – PAGE 1 Executive Dashboard

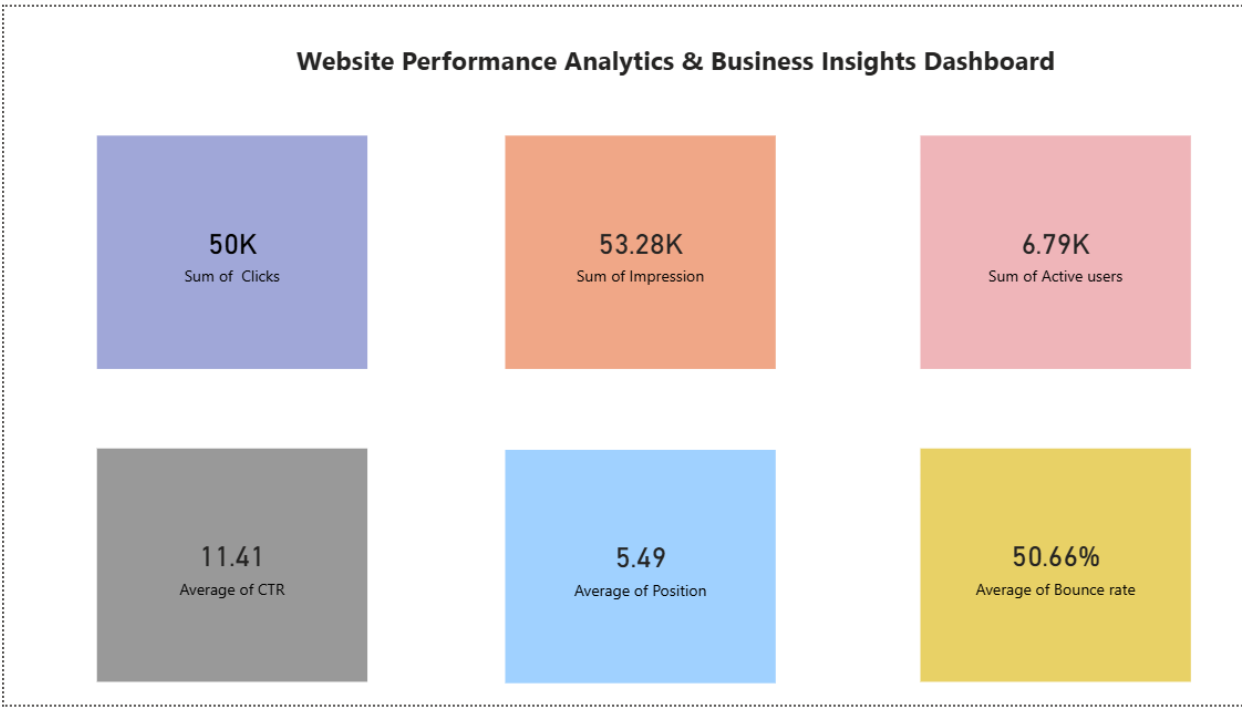


Figure 1: Executive Dashboard – Website Performance Analytics & Business Insights Dashboard

What This Page Shows

The Executive Dashboard is the first page of the Power BI report. It displays six large KPI (Key Performance Indicator) cards that give a complete summary of TACTIVE's website performance at a single glance. Each colored card shows one important metric.

KPI Card	Color	Value (Combined)	What It Means
Sum of Clicks	Purple	50K (50,000)	Total users who visited the site from Google
Sum of Impressions	Orange	53.28K	Times the site appeared in Google search results
Sum of Active Users	Pink	6.79K	Unique users who actively engaged with the site
Average CTR	Gray	11.41	% of people who clicked after seeing the site
Average Position	Light Blue	5.49	Average Google search ranking position
Average Bounce Rate	Yellow	50.66%	% of users who left after seeing only one page

These six cards represent the combined view across both April and May 2026. An average CTR of 11.41% and a combined 50,000 clicks shows strong organic search performance for a construction company.

CHAPTER 3: DASHBOARD – PAGE 2 Monthly Website Performance Analysis

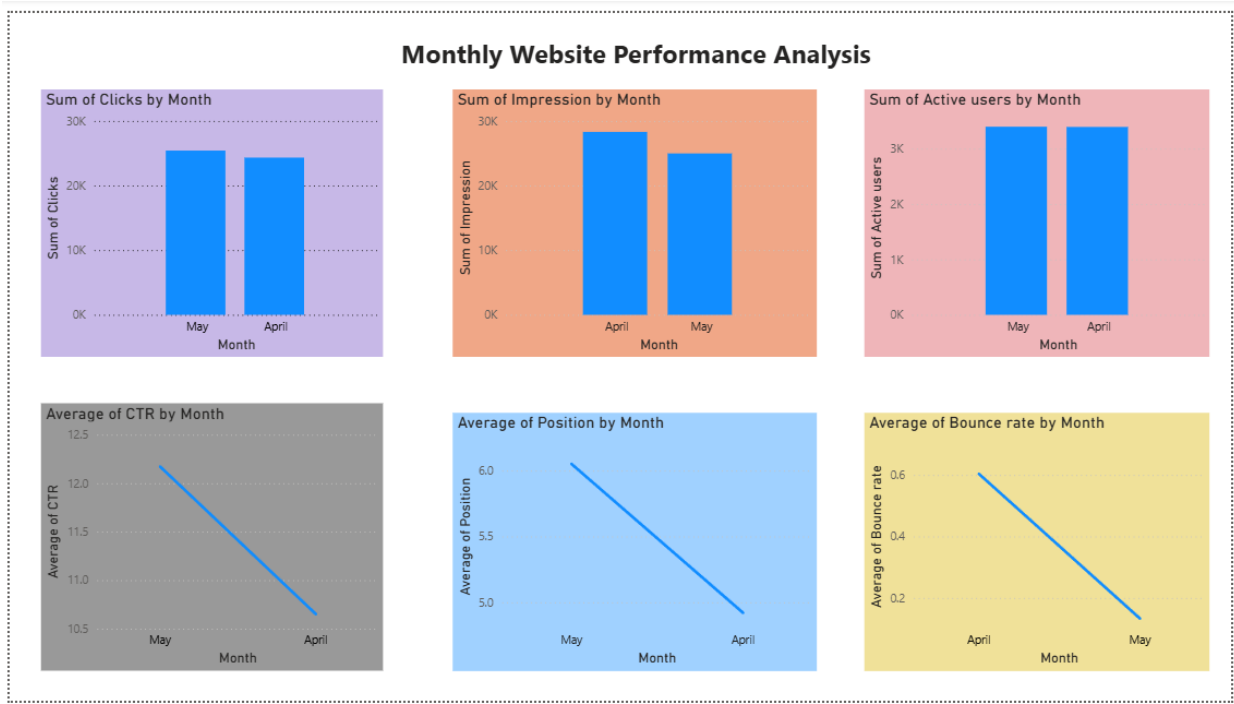


Figure 2: Monthly Website Performance Analysis – April vs May 2026

What This Page Shows

This page compares April and May 2026 side by side using six charts — three bar charts (top row) and three line charts (bottom row). Each chart focuses on one metric and shows how it changed between the two months.

Chart-by-Chart Explanation

- Sum of Clicks by Month (Top Left – Purple): May recorded slightly more clicks (~25K) than April (~24K). This shows traffic grew month over month.
- Sum of Impressions by Month (Top Middle – Orange): April had more impressions (~28K) than May (~25K). This means the site appeared in search results more often in April.
- Sum of Active Users by Month (Top Right – Pink): Both months had nearly the same number of active users (~3,400), showing a stable audience.
- Average CTR by Month (Bottom Left – Gray): CTR dropped from ~12% in May to ~10.6% in April, shown as a downward line. May had better CTR.
- Average Position by Month (Bottom Middle – Blue): Position declined from ~5 in May to ~6 in April. In SEO, a lower number is better, so May had a slightly worse position.
- Average Bounce Rate by Month (Bottom Right – Yellow): Bounce rate was 60% in April and fell to 13% in May — the most dramatic positive change in the entire dashboard.

CHAPTER 4: DASHBOARD – PAGE 3 Top Pages Performance Analysis

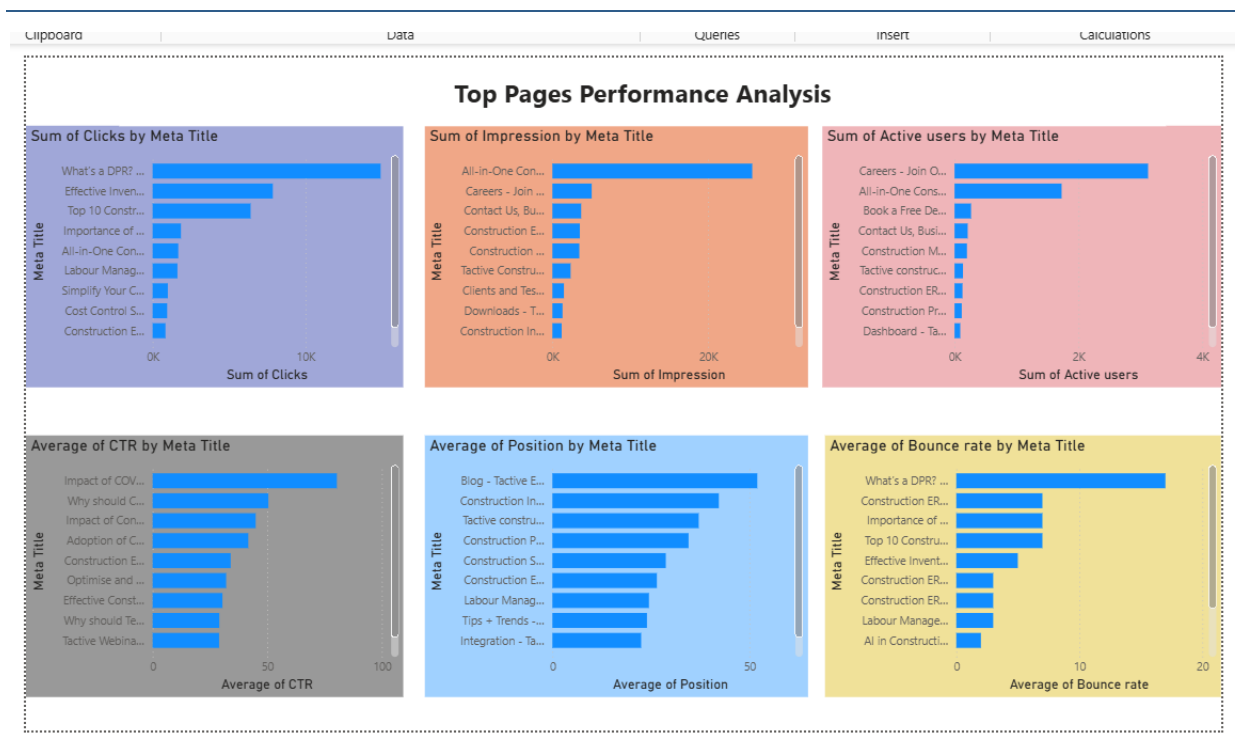


Figure 3: Top Pages Performance Analysis – Performance by Individual Page (Meta Title)

What This Page Shows

This page breaks down performance by individual web pages (identified by their Meta Title). Six bar charts show which pages performed best on each KPI. This helps the team understand which content is working well and which needs improvement.

Key Findings from Each Chart

- Sum of Clicks by Meta Title (Top Left): The page 'What's a DPR?' received the highest clicks, followed by 'Effective Inventory' and 'Top 10 Construction'. These are the most traffic-generating pages.
- Sum of Impressions by Meta Title (Top Middle): The 'All-in-One Construction ERP Software' page appeared most in search results, followed by Careers and Contact Us pages.
- Sum of Active Users by Meta Title (Top Right): The Careers page led in active users (~3,500–4,000), suggesting high engagement from job seekers visiting the site.
- Average CTR by Meta Title (Bottom Left): Pages like 'Impact of COVID', 'Why should C...' and 'Impact of Construction' showed the highest CTR percentages (above 50–100).

- Average Position by Meta Title (Bottom Middle): The Blog page ranked best in search results. Construction-related pages ranked around positions 30–50.
- Average Bounce Rate by Meta Title (Bottom Right): 'What's a DPR?' had the highest bounce rate (~18), suggesting visitors leave quickly after reading that page.

CHAPTER 5: DASHBOARD – PAGE 4 Page Performance Summary

Month	Meta Title	Sum of Clicks	Sum of Active users
April	A multi-dimensional construction ERP for the modern builder - Tactive	3	1.00
April	Adoption of Construction Management Software to Achieve Cost Control.	297	1.00
April	AI in Construction Industry Artificial Intelligence for Construction	0	0.00
April	All-in-One Construction ERP Software	890	864.00
April	Application of an Automated Inventory Tracking Alert System for URCC	59	1.00
April	Automating the Labor Billing System for URC - Tactive ERP	140	1.00
April	Bank Transfer Issues - Tactive ERP	17	1.00
April	Benefits of construction ERP in managing Labour, Machinery and Plant?	60	0.00
April	Benefits of Construction ERP Software by Role - Tactive ERP	49	0.00
April	Benefits of Managing Finance & Accounts using Construction ERP?	41	0.00
April	Big5 Digital Festival 2020 - Redefining Construction - Tactive ERP	0	0.00
April	BIM, IoT, & ERP Software - Future of Construction Technology - Tactive	49	0.00
April	Blog - Tactive ERP	34	1.57
April	Book a Free Demo Tactive ERP – Construction Management Software	0	133.00
April	Brochure Downloads - Tactive ERP	1	7.00
April	Careers - Join Our Team at Tactive Construction ERP Software	124	1,562.00
April	Case Studies - Tactive ERP	0	5.00
April	Challenges in Implementing Construction ERP System - Tactive ERP	94	1.00
April	Clients and Testimonials Tactive Software Systems Private Limited	6	41.00
April	Construction Accounting Software Finance and Accounts Management ERP	3	13.00
April	Construction Equipment Management Software Tactive ERP	0	16.00
April	Construction Equipment Management Solution - Tactive ERP	70	2.00
April	Construction ERP - Inventory Management Software - Tactive ERP	115	1.00
April	Construction ERP Software Tactive Software Systems Private Limited	13	64.00
April	Construction ERP Software for Construction Procurement Managers.	105	0.00
April	Construction ERP Software for Technology Integration and Coordination	15	0.00
April	Construction ERP software Implementation Guide Tactive ERP	344	1.00
April	Construction ERP Software Modules Tactive ERP Solutions	413	1.50
Total		24327	3,391.69

Figure 4: Page Performance Summary – Detailed Clicks and Active Users per Page

What This Page Shows

This page is a detailed data table listing every page of the TACTIVE website with its performance data for April. Each row shows the Month, the page's Meta Title, the total Clicks, and the number of Active Users for that page.

Important Observations

- The table shows April data with a grand total of 24,327 clicks and 3,391.69 active users at the bottom.
- The 'All-in-One Construction ERP Software' page was the strongest performer with 890 clicks and 864 active users — showing it attracted highly engaged visitors.
- The 'Careers – Join Our Team' page had 124 clicks but 1,562 active users, the highest active user count. This likely reflects a page that users revisit or spend extended time on.
- Several pages such as 'AI in Construction Industry' had 0 clicks, indicating they are either new pages or not yet ranking in Google.
- Pages like 'Book a Free Demo' had 0 clicks but 133 active users, showing that users reach this page via internal links rather than search.

This table is valuable for identifying both high performers and underperforming pages that need SEO attention.

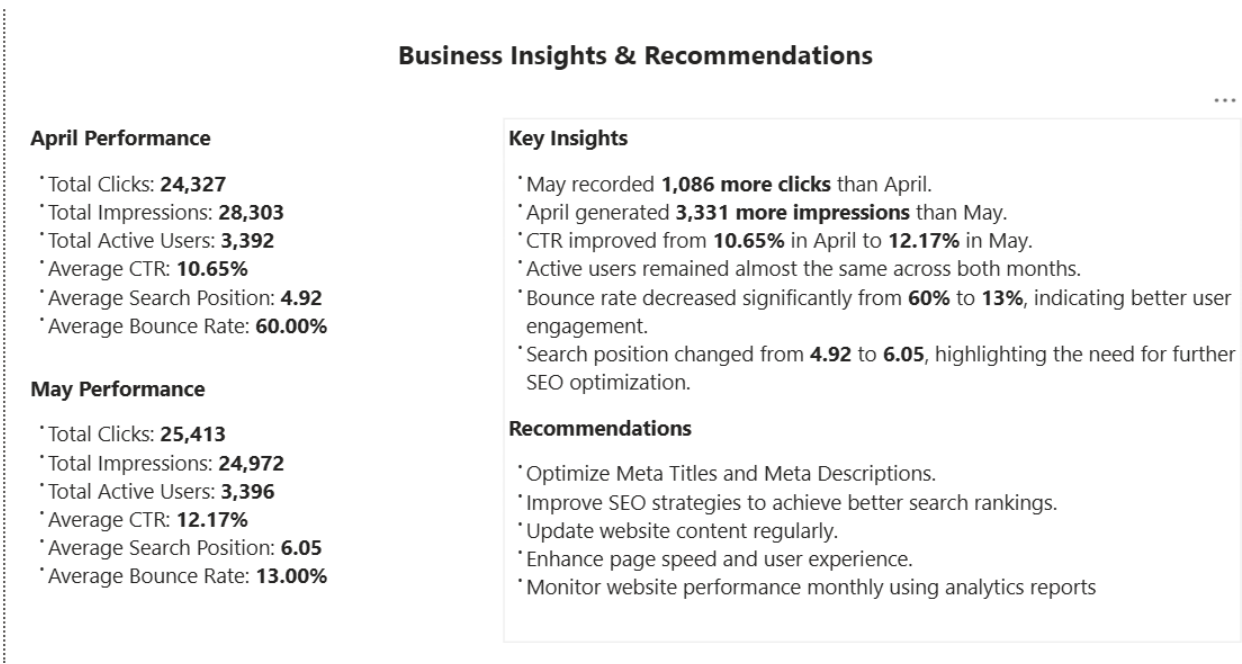


Figure 5: Business Insights & Recommendations – Summary and Action Points

What This Page Shows

The final dashboard page presents the key business findings and recommended actions in plain language. It is divided into three sections: April Performance, May Performance, and Key Insights with Recommendations. This page is designed for the TACTIVE management team to read and act upon directly.

April vs May Performance Summary

Metric	April 2026	May 2026
Total Clicks	24,327	25,413
Total Impressions	28,303	24,972
Total Active Users	3,392	3,396
Average CTR	10.65%	12.17%
Average Search Position	4.92	6.05
Average Bounce Rate	60.00%	13.00%

Key Insights (from dashboard)

- May recorded 1,086 more clicks than April — traffic grew.
- April had 3,331 more impressions than May — but fewer clicks were converted in April.
- CTR improved from 10.65% to 12.17% — the site listing became more compelling.
- Active users remained nearly the same across both months.
- Bounce rate fell dramatically from 60% to 13% — users engaged much more in May.
- Search position changed from 4.92 to 6.05 — slight decline, needs improvement.

Recommendations (from dashboard)

- Optimize Meta Titles and Meta Descriptions to improve CTR further.
- Improve SEO strategies to recover and improve the search position back below 5.

- Update website content regularly to stay relevant in Google search results.
- Enhance page speed and user experience to keep bounce rate low.
- Monitor website performance monthly using analytics reports.

CHAPTER 7: RESULTS, CONCLUSION & FUTURE SCOPE

7.1 Key Results

Metric	April	May	Change
Clicks	24,327	25,413	+4.46% ✓
Impressions	28,303	24,972	-11.77%
CTR	10.65%	12.17%	+1.52 pp ✓
Avg. Position	4.92	6.05	Declined ↓
Active Users	3,392	3,396	Stable →
Bounce Rate	60%	13%	-47 pp ✓✓

7.2 Conclusion

This internship project successfully developed a five-page SEO Analytics Dashboard using Power BI for TACTIVE Building Construction. The dashboard provides a clear and visual way to monitor website performance using real data from Google Search Console and Google Analytics.

The analysis shows that the TACTIVE website improved significantly in user engagement between April and May 2026. The Bounce Rate dropped from 60% to 13% and CTR improved from 10.65% to 12.17%, both strong positive signs. The average search position declined slightly from 4.92 to 6.05, which requires attention going forward.

This internship provided valuable hands-on experience in data collection, cleaning, Power BI dashboard development, and deriving business insights — all essential skills for a career in Data Science and Business Intelligence.

7.3 Future Scope

- Extend the dashboard to cover 12 months of data for seasonal trend analysis.
- Connect Power BI to live Google Search Console data for real-time monitoring.
- Add conversion tracking (form submissions, calls) to measure lead generation.
- Use AI forecasting in Power BI to predict future clicks and impressions.

References

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